

Project Title : EquiTrack: Comprehensive Web Platform for Stock Tracking and Portfolio Watchlist Management

Problem Statement

Beginner and retail investors often find comprehensive stock trading platforms overwhelming, cluttered with excessive financial jargon, or locked behind expensive subscriptions. Conversely, basic financial sites lack organized watchlist capabilities and clear visual metrics. There is a need for a streamlined, accessible, and responsive web platform that provides essential stock visualisations, curated watchlists, and trending market insights in a single unified dashboard.

Objectives

- Provide an intuitive visual dashboard displaying market trends, top gainers, top losers, and suggested stocks.
- Enable secure authentication allowing users to build and persist custom watchlists across sessions.
- Deliver interactive, multi-timeframe price charts (1 Day, 1 Week, 1 Month, 1 Year) for in-depth analysis.
- Deliver fast search capability by company name or stock symbol with detailed financial metadata pages.

Existing System

Existing stock platforms generally fall into two categories: high-end professional tools (like Bloomberg Terminal or TradingView) which have steep learning curves and overwhelming interfaces, or basic market news sites that require repetitive search queries and do not persist personalized user watchlists. Most existing free tools are cluttered with ads, hindering quick stock analysis.

Proposed System

EquiTrack provides a streamlined web application focused on clarity, accessibility, and speed. Users create an account to maintain their own custom watchlist across devices. The dashboard provides a snapshot of current market dynamics (top gainers/losers, trending suggestions) while company detail pages give detailed stock performance metrics paired with interactive charts. The responsive interface removes noise, focusing on data readability and quick interaction.

Scope

- **In Scope:** User registration, secure login/logout, interactive dashboard, live market top gainers/losers, stock symbol search, company overview pages, multi-timeframe interactive price charts, personalized watchlist creation/modification, and user profile management.

- **Out of Scope:** Real-time automated stock trading/execution, brokerage account integrations, fiat currency deposit/withdrawal, and live crypto asset tracking.

Expected Users

- **Retail Investors:** Individuals seeking a clean, fast dashboard to monitor selected stocks and view market trends.
- **Beginner Investors:** Users learning to analyze stock price movements over varied timeframes without being overwhelmed by advanced trading controls.
- **System Administrator:** Admin users overseeing account management and monitoring system performance.

Expected Features

- **User Registration & Authentication:** Secure sign-up/login with email verification and duplicate registration prevention.
- **Market Dashboard:** Centralized dashboard highlighting market trends, top gainers, top losers, and curated stock suggestions.
- **Stock Search & Company Details:** Search by company name or ticker symbol to view current price, daily highs/lows, description, and charts.
- **Personalized Watchlist:** One-click option to save or remove stocks to/from a persistent personal watchlist.
- **Interactive Stock Charts:** Customizable price charts with switchable time horizons (1 Day, 1 Week, 1 Month, 1 Year).
- **Profile Management:** Capability to view and update account profiles.

Technology Stack

Layer	Technology Chosen	Purpose
Frontend	React.js, HTML5, CSS3, Tailwind CSS	Build a responsive, modern user interface with dynamic component re-rendering for real-time charting.
Data Visualization	Chart.js / Recharts	Render smooth, interactive stock price charts across multiple timeframes.

Backend	Node.js with Express.js	Provide RESTful APIs for user authentication, market data retrieval, and watchlist database operations.
Database	MongoDB / PostgreSQL	Store user credentials securely and maintain custom user watchlist collections.
External APIs	Finnhub API / Alpha Vantage API	Fetch real-time stock quotes, company profiles, and historical market data.